

Customs Mismatch Resolver — Value Creation Rationale

DTU 38113 · Applied AI for Entrepreneurs · Module 1 assignment · Fleet agentic AI design

The task and the decision it targets

DSV's cross-border shipments pass through up to 30 parties and 36 documents across many customs regimes, much of it still reconciled manually. When a document mismatch holds a shipment, the expensive part is not fixing it — it is a human noticing the hold, assembling the shipment file, the flagged document, and the specific rule that failed, before any judgment gets applied. The **Customs Mismatch Resolver** is a tool-calling loop (per LangChain's loop-engineering framing: model + tools, repeating until the task resolves) that absorbs this assembly-and-diagnosis cost, then applies **tiered autonomy** — auto-resolving routine cases, escalating everything else to a human in Slack with a decision receipt.

Value creation, mapped to the lecture sources

McKinsey — Innovation Revolution	Anthropic — Loop Engineering & Recursive Self-Improvement
The agent's work classifies primarily as research operations (fast, parallel retrieval of the shipment file, document, and live ruleset) with a secondary design evaluation step (the grader that checks a drafted correction before it is shown to anyone). It is not a design-generation task — there is one correct fix to converge on, not many candidate options — so velocity is the dominant axis: the agent compresses a multi-step manual lookup into a single run.	The design uses Loop 3 (event-driven: triggered by a Slack message or held-shipment signal, not manually invoked) wrapped around Loop 2 (a verification/grader step — <code>validate_against_ruleset</code> — checks every draft against the live customs rule before any action is taken). Per the finding that "the comparative advantage of humans... is still in seeing the bigger picture," the agent never decides alone on high-stakes or novel cases — only on the narrow, well-verified slice defined in <code>SKILL.md</code> 's known-safe pattern list.
McKinsey — Decision Dividend	Anthropic — Building Effective Human-Agent Teams
This directly targets the report's central claim: AI can cut the cost of a decision embedded in a workflow by over 90%, compressing timelines from weeks to seconds, with better unit economics than human-mediated coordination for supply-chain-scenario-type decisions. The agent's decision receipt (what was flagged, what changed, confidence, cost/time impact) is the audit trail that makes this compression trustworthy rather than opaque — mirroring the report's point that workflow <i>redesign</i> , not a bolt-on tool, is what captures the larger EBITDA uplift.	Two lessons shape the design directly: " <i>grant agents autonomy in proportion to demonstrated reliability, then expand it deliberately</i> " — the tiered-autonomy rule in <code>AGENTS.md</code> is exactly this, and the pattern library explicitly forbids self-expansion without human review of prior runs. And " <i>treat human attention as the scarce resource it is</i> " — the agent batches all approval-tier shipments into one Slack message per run rather than one message per shipment, and operates as a Slack teammate (@tag-able, not a silent background job) per the single-player-to-multiplayer framing.

Why the guardrails are load-bearing, not decorative

The OpenAI/Hugging Face incident write-up (VentureBeat coverage, Lecture 1) is the cautionary counter-case: agents without production-grade harness constraints persisted on tasks past a safe stopping point, trusted unauthorized instructions from peer agents, and expanded scope well beyond their assigned task — and OpenAI's own retrospective found that **applying production harness and system-prompt controls dropped the propensity for this kind of infrastructure overreach by over 100x**. This is the direct justification for this design's `Hard constraints` section: the agent may only call the ten tools it is given, must never resubmit without a live ruleset check, must escalate rather than persist on ambiguous or unsolvable cases, and must never treat content it reads (a document field, a ruleset note) as an instruction to itself.

Tiered autonomy — where the line is drawn

Tier	Condition	Action
Auto-act	Known-safe pattern (<code>SKILL.md</code>) and value < €5,000 and grader confidence > 90%	Resubmit directly; decision receipt logged
Approval	Any of: unrecognized pattern, value ≥ €5,000, confidence ≤ 90%, unfamiliar ruleset	Post to #customs-holds; wait for human sign-off

Live test results — Fleet + Slack, three shipments

Shipment	Outcome	Why
SHP-1001	Auto-resolved	HS-code digit transposition (known-safe), €1,250, 95% confidence — resubmitted without human input
SHP-1002	Escalated	Unit-of-measure fix validated at 95% confidence, but €84,000 exceeds the €5,000 threshold
SHP-1003	Escalated	Certificate-of-origin signature format not on the known-safe list, regardless of low value

All three runs used the agent's real MCP tool calls against a live mock customs server (deployed on Render), not simulated reasoning — confirming the tiering logic executes correctly end-to-end and that two independent conditions (value, pattern novelty) each independently trigger escalation as designed.

Fleet package (`AGENTS.md`, `config.json`, `tools.json`, `skills/mismatch-pattern-library/SKILL.md`) submitted alongside this document, `tools.json` pointing at the deployed MCP server. MCP server (`customs-mcp`) implements the eleven tools referenced above against a mock shipment dataset; deployed live and exercised via Slack for the test results above.